

EMBED: FOCUS + CONTEXT

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The Big Picture



→ Embed

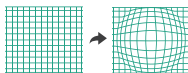
→ Elide Data



→ Superimpose Layer



→ Distort Geometry



Reduce

→ Filter



→ Aggregate



→ Embed





Why Embed?

Why Embed?



- The goal of embedding focus and context together is to mitigate the potential for disorientation that comes with standard navigation techniques such as geometric zooming
- With realistic camera motion, only a small part of world space is visible in the image when the camera is zoomed in to see details for a small region



Elide

Elide

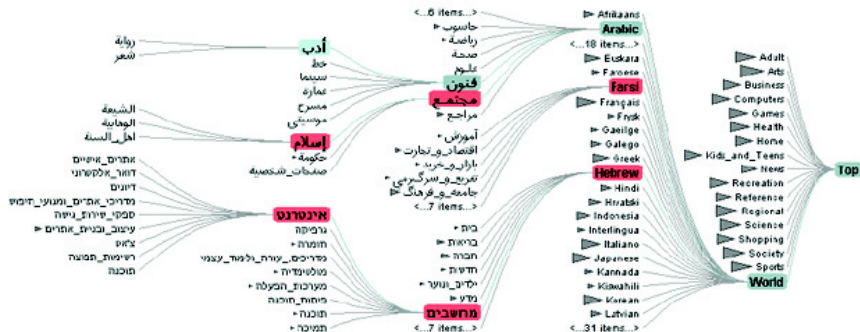


- One design choice for embedding is **elision**: some items are omitted from the view completely, in a form of dynamic filtering.
- Other items are summarized using dynamic aggregation for context, and only the focus items are shown in detail.
- A general framework for reasoning about these three sets of items is a degree of interest (*DOI*) function:

$$DOI = I(x) - D(x, y).$$

In this equation, I is an interest function; D is the distance, either semantic or spatial; x is the location of an item; y is the current focus point

- DOITrees Revisited uses elision to show multiple focus nodes within context in a 600,000 node tree



Idiom



System	DOITrees Revisited
What: Data	Tree.
How: Encode	Node–link layout.
How: Reduce	Embed: elide, multiple foci.
Scale	Nodes: hundreds of thousands.



Superimpose

Superimpose

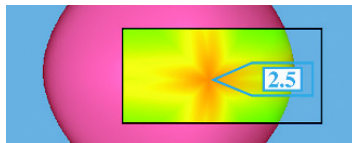


- Another choice for integrating focus and context is the use of superimposed layers. In this case, the focus layer is limited to a local region, rather than being a global layer that stretches across the entire view to cover everything



Example: Toolglass and Magic Lenses

- The Toolglass and Magic Lenses idiom provides focus and context through a superimposed local layer: the see-through lens color codes the patchwork sphere with Gaussian curvature information and provides a numeric value for the point at the center.



Idiom



System	Toolglass and Magic Lenses
What: Data	Spatial, quantitative curvature attribute across surface.
How: Encode	Use given, color by curvature.
How: Reduce	Embed: superimpose.



Distort

Distort



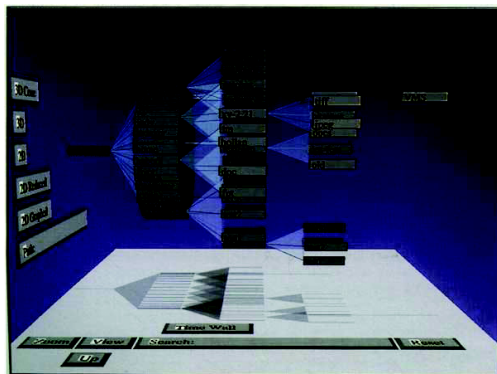
In contrast to using elision or layers, many focus+context idioms solve the problem of integrating focus and context into a single view using geometric distortion of the contextual regions to make room for the details in the focus regions.

1. There is the choice of the number of focus regions: is there only a single region of focus, or does the idiom allow multiple foci?
2. Another choice is the shape of the focus: is it a radial, rectangular, or a completely arbitrary shape?
3. A third choice is the extent of the focus: is it global across the entire image, or constrained to just a local region?
4. A fourth choice is the interaction metaphor.



Example: 3D Perspective

- The Cone Tree system used 3D perspective for focus+context, providing a global distortion region with a single focus point, and using standard geometric navigation for interaction.



Idiom



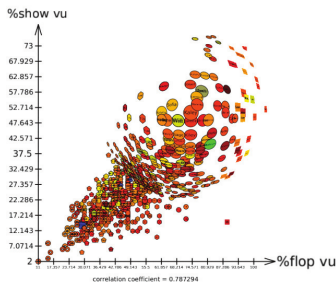
System	Cone Trees
What: Data	Tree.
How: Encode	3D node–link layout.
How: Reduce	Embed: distort with 3D perspective; single global focus region.
Scale	Nodes: thousands.



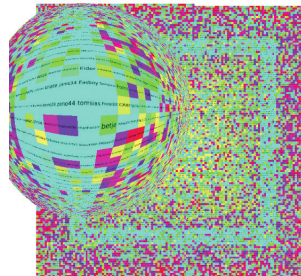
Example: Fisheye Lens

Focus+context with interactive fisheye lens, with poker player dataset.

- (a) Scatterplot showing correlation between two strategies.
- (b) Dense matrix view showing correlation between a specific complex strategy and the player's winning rate, encoded by color.



(a)



(b)

Idiom

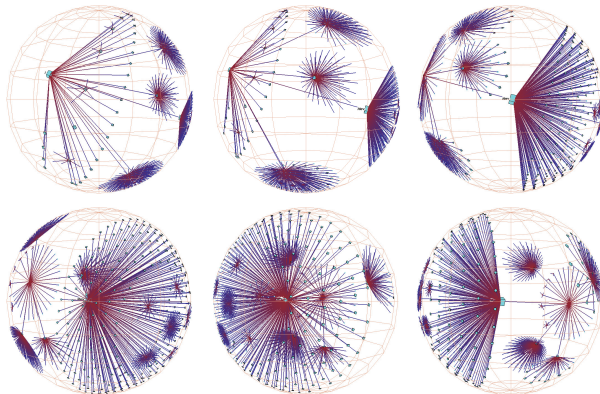


Idiom	Fisheye Lens
What: Data How: Encode How: Reduce	Any data. Any layout. Embed: distort with fisheye; single focus, local radial region, moveable lens interaction metaphor.



Example: Hyperbolic Geometry

- Animated transition showing navigation through 3D hyperbolic geometry for a file system tree laid out with the H3 idiom, where the first three frames show hyperbolic translation changing the focus point and the last three show standard 3D rotation spinning the structure around.



Idiom



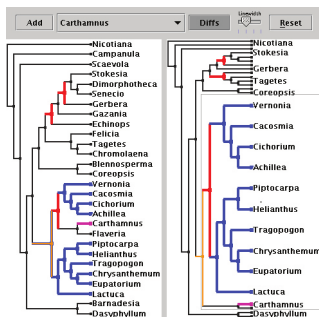
Idiom	Hyperbolic Geometry
What: Data How: Encode How: Reduce	Tree or network. Hyperbolic layout. Embed: distort by projecting from hyperbolic to Euclidean space; single global radial focus; hyperbolic translation interaction metaphor.



Example: Stretch and Squish Navigation

TreeJuxtaposer uses stretch and squish navigation with multiple rectangular foci for exploring phylogenetic trees.

- (a) Stretching a single region when comparing two small trees.
- (b) Stretching multiple regions within a large tree.



(a)



(b)

Example: Stretch and Squish Navigation (cont.)



PRISquenceJuxtaposer supports comparing gene sequences using the stretch and squish navigation idiom with the guaranteed visibility of marks representing items with a high importance value, via a rendering algorithm with custom subpixel aggregation.



Idiom

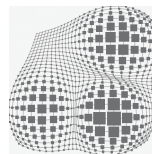
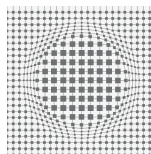
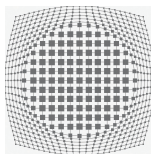


Idiom	Stretch and Squish Navigation
What: Data How: Encode How: Reduce	Any data. Any layout. Embed: distort with stretch and squish; multiple foci, global rectangular regions, stretch and squish navigation interaction metaphor.

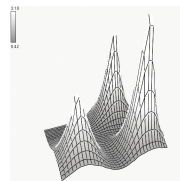
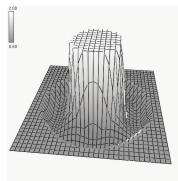
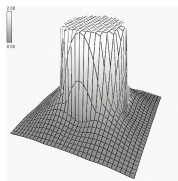


Example: Nonlinear Magnification Fields

- General frameworks calculate the magnification and minimization fields needed to achieve desired transformations in the image. (a) Desired transformations. (b) Calculated magnification fields.



(a)



(b)

Idiom



Idiom	Nonlinear Magnification Fields
What: Data How: Encode How: Reduce	Any data. Any layout. Embed: distort with magnification fields; multiple foci, local arbitrary regions, lens or stretch or data driven interaction metaphors.



Costs and Benefits: Distortion



Costs and Benefits: Distortion

- One cost is the problem that distance or length judgements are severely impaired, so distortion is a poor match with any tasks that require such comparisons
- One potential cost is that users may not be aware of the distortion, and thus misunderstand the underlying object structure
- Another cost is the internal overhead of maintaining **object constancy**, which is the understanding that an item seen in two different frames represents the same object, just seen from a different viewpoint

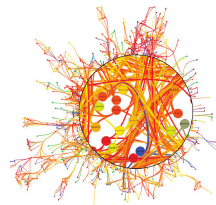


Example

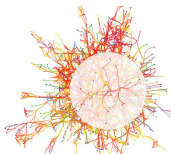
- Four approaches to graph exploration. (a) Fisheye lens. (b) Magnifying lens. (c) Neighborhood highlighting with layering. (d) Neighborhood highlighting with both layering and Bring and Go interaction



(a)



(b)



(c)



(d)

References



Goodfellow, I., Bengio, Y., and Courville, A. (2016).

Deep learning.

MIT press.



Munzner, T. (2014).

Visualization analysis and design.

CRC press.



Russell, S. and Norvig, P. (2016).

Artificial intelligence: a modern approach.

Pearson Education Limited.



Ward, M. O., Grinstein, G., and Keim, D. (2015).

Interactive data visualization: foundations, techniques, and applications.

CRC Press.